

# Certified Correction of Vision–Language Hallucinations: Guarantees, Gains and Failure Modes under a Pre-specified Protocol

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## Abstract

Selective prediction gives a vision–language model (VLM) a distribution-free guarantee on the answers it emits and pays for it by abstaining on the rest. When the base error rate  $\mu$  exceeds the target  $\alpha$ , any predictor that may only emit or abstain the model’s own answer must abstain on at least  $(\mu - \alpha)/(1 - \alpha)$  of inputs. We ask what changes when the predictor may emit a different answer. Certified-Correction Risk Control (CCRC) accepts the model’s answer where a correctness score is high, substitutes the answer of a distinct evidence source (here an open-vocabulary detector) where that source is decisive, abstains otherwise, and certifies the error rate of everything emitted with a fixed-sequence Clopper–Pearson test.

We prove validity for policy families indexed by pre-specified score values under i.i.d. sampling of calibration units, and we evaluate CCRC under a protocol built so that the proved procedure and the evaluated procedure coincide: folds are unions of whole images, thresholds and repair gate are values fixed before calibration, the start of the test sequence is derived analytically, and the repair gate is fixed on external data, chosen on the fit fold, or certified by a union bound. On five benchmark–backbone cells, certified coverage over filtering changes by +0.8 to +7.9 points with an externally fixed gate, positive in all ten (cell,  $\alpha$ ) combinations; an outer image-level bootstrap excludes zero only on the largest cell (250 images). Most of the repaired mass re-emits the model’s own answer with detector confirmation, and only 0.04–1.7% of test items have their answer changed; no false-positive existence error is corrected. The correctness score, not the repair, supplies most of the coverage, and fixed-sequence testing started at the minimal certifiable emitted set is fragile with small calibration folds. A matched-prefix experiment shows that mid-caption repair increases downstream hallucination, which blocks a fixed-point route to a sequential guarantee.

## 1 Introduction

A vision–language model that answers “is there a mouse in the image?” wrongly is more useful if it says so. Selective prediction does this: a score ranks the model’s answers, a threshold is calibrated on held-out data, and the model answers only above it. Conformal and risk-controlling variants [2, 3, 7] certify, with a finite-sample guarantee that makes no assumption on the score or the model, that the error rate among the answers that are emitted stays below a target  $\alpha$ .

The guarantee is paid for in coverage, and part of the bill cannot be reduced. Kotte [33] shows that when the base error rate  $\mu$  exceeds  $\alpha$ , any predictor that may only emit or abstain the model’s own answer must abstain on at least  $(\mu - \alpha)/(1 - \alpha)$  of inputs, whatever the score,

the concentration inequality or the calibration budget (Proposition 1 below). The proof uses one premise: an abstained item is an item whose model answer stays wrong. A predictor that may emit a *different* answer is outside the premise.

This paper studies that possibility in the simplest setting where it can be made exact: yes/no object-existence questions, with an open-vocabulary detector as a second, distinct source of evidence. *Certified-Correction Risk Control* (CCRC) accepts the model’s answer where a correctness score is high, replaces it with the detector’s answer where the detector’s evidence is decisive, abstains otherwise, and certifies the error rate of the whole emitted set, accepted and substituted answers together.

**What the paper contributes.** Our aim is a result that survives a strict reading of the protocol. Four design decisions follow from that aim, and each is a contribution in its own right.

1. *The proved procedure is the evaluated procedure.* Theorem 1 covers policy families indexed by threshold *values* fixed before calibration under i.i.d. sampling of calibration units. Every reported number uses such a family: thresholds and repair gate are read off the fit fold, never off the calibration sample whose errors are counted.
2. *The sampling unit is stated and respected.* POPE contributes six questions per image; folds here are unions of whole images, we measure the within-image dependence of errors, and a known-population simulation shows how clustering would degrade the guarantee and how a one-item-per-image calibration restores it.
3. *The repair gate is not tuned on the evaluation data.* We report three ways of fixing it that keep the guarantee: a constant chosen on a different benchmark, a choice made on the fit fold, and a union bound over a pre-specified set of gates.
4. *Uncertainty is stated at the level of the data.* Gains are accompanied by an outer image-level bootstrap, because hundreds of random splits of one benchmark measure the procedure’s sensitivity to the split, not the benchmark’s sampling error.

**What we find.**

- With the gate fixed on an external benchmark, CCRC certifies +0.8 to +7.9 points more coverage than filtering in all ten (cell,  $\alpha$ ) combinations, and the image-bootstrap interval excludes zero only on the largest cell (250 images). On the others the sign is right in 65–95% of bootstrap replicates but cannot be called (§6.1).
- Choosing the gate on the fit fold, or certifying three gates by a union bound, is valid but costs more than it returns when the calibration fold holds 150–200 items (−6.2 to +9.4 and −16.5 to +4.4 points over all cells); with 500 items both help.
- The repair is mostly confirmation. Between 41 and 96% of repaired items are items on which the detector agrees with the model, and the answer is changed for only 0.04–1.7% of test items. The one-directional gate never corrects a false-positive existence error, and a polarity-symmetric gate does not either (§6.3).

- The start of the fixed sequence matters more than the repair on small folds. Starting where a single error can be tolerated ( $k \geq 38$  at  $\alpha = \delta = 0.10$ ) rather than where none can ( $k \geq 22$ ) raises filtering coverage on the smallest cell (calibration folds of 150 items) from 22% to 41% when the first hypothesis sits at the minimal count, and removes the apparent loss on AMBER that the conventional start produces (§6.4).
- Certified coverage comes mostly from the score: a detection-grounded score certifies 68% on POPE-1500 against 41% for model confidence and 0.5% for the image–text similarity score of ConflVLM, while certified repair adds 3–6 points (§6.5).
- In free-form generation, repairing a claim mid-caption increases downstream hallucination in a matched-prefix experiment (§8), which blocks the fixed-point argument we had hoped to use for a sequential guarantee.

**Scope.** Everything here concerns binary existence questions, one detector, one family of backbones and benchmarks whose images overlap (642 distinct images across all cells, and the POPE-adversarial cells are subsets of the POPE-1500 cell’s 250 images). We do not claim that correction is generally useful, only that under a protocol whose guarantee we can state it is valid and its gain is measured.

## 2 Related work

The survey covers work available up to 4 October 2026; claims of the form “no method does X” refer to the literature we surveyed up to that date and are not claims about the field.

**Hallucination in vision–language models.** Multimodal hallucination is usually split into existence, attribute and relation errors [63]; surveys [5, 29, 45] attribute it to language-prior dominance, weak visual attention, instruction-tuning artefacts [43] and miscalibration [26]. Existence claims can be checked by an open-vocabulary detector [46, 48]; attribute and relation claims need finer perception than a detector supplies, which is why the benchmark’s composition, and not only its difficulty, determines whether our evidence channel applies (§6.7). POPE [40] probes existence with balanced yes/no questions under random, popular and adversarial negative sampling; AMBER [63] adds an LLM-free discriminative split; MME [21], GQA [28] and HallusionBench [24] are used here as settings where the detector mostly does not apply.

Mitigation methods modify decoding [10, 20, 27, 39, 74, 75], intervene on heads or positional structure [11, 16, 30, 62, 76], retrain [43, 58, 73], or verify and rewrite after generation [15, 36, 68, 69, 72, 77]. Post-hoc verification performs the same three steps as CCRC (flag, consult external evidence, substitute) and none of these systems certifies the substitution. Lowering the base risk  $\mu$  lowers the abstention floor, so these methods compose with ours; §6.5 runs CCRC on top of visual contrastive decoding.

**Conformal prediction and distribution-free risk control.** Conformal prediction [1, 38, 51, 60] gives finite-sample coverage under exchangeability. Risk-controlling prediction sets [7] and conformal risk control [2] control the expectation of a monotone loss; Learn-then-Test [3]

treats the choice among configurations as multiple testing and admits fixed-sequence testing over a pre-ordered family at full  $\delta$ , which we use. For a binary loss the exact Clopper–Pearson bound [13] is tighter than Hoeffding, empirical Bernstein or betting bounds [67]. Selective prediction with abstention predates the conformal treatment [6, 12, 14, 17, 22]; the floor of Proposition 1 concerns the action space and applies to these formulations too.

**Conformal methods for language and vision–language models.** Conformal language modelling [53] calibrates sampling, stopping and rejection; conformal factuality [50] backs off to a subset of claims; coherent factuality [57] filters whole reasoning steps; adaptive variants improve conditional coverage of the filter [56]; and a robustness audit shows that claim filtering for retrieval-augmented models becomes vacuous at high factuality levels and is fragile to shift [9]. Conformal alignment [25], conformal multiple choice [35] and calibrated help-seeking [55] certify act-or-defer decisions. In the multimodal setting ConflVLM [41] filters unsupported claims scored by image–text similarity, Wagner et al. [61] study abstention along correctness and answerability, and IntroConformal [4] builds conformity scores from the model’s hidden states and self-verification, without external verifiers. Inference-time conformal reasoning [65] moves calibration inside generation and intervenes by truncating. All of these filter, back off, stop or abstain. One method certifies a revised output: conformal linguistic calibration [31] replaces claims by less specific ones with a guarantee on the rewritten text; it is text-only and the rewrite generalises a claim rather than substituting externally verified evidence.

**Uncertainty estimation without a guarantee.** Our correctness score belongs to the literature on generative uncertainty: self-evaluation [32], semantic entropy [19, 34], verbalised confidence [42], sampling consistency [47, 66], and token-level conformal variants [8, 70]. These scores interrogate the model about its own output; detector and similarity scores are perceptual evidence from a distinct predictor.

**The closest works.** Kotte [33] proves the abstention floor for text-only structured generation and recommends relaxing  $\alpha$ ; the floor admits three responses: relax the target, lower  $\mu$  or sharpen the score, or change the action space. We take the third. BCEA [71] sharpens the score by spending a visual budget to re-read the image and recalibrating on post-acquisition scores; the answer is still the model’s own. Its current version certifies a Bonferroni-corrected Clopper–Pearson test over a threshold grid fixed independently of the calibration sample, and labels scanning the calibration scores at full confidence an uncertified practical variant. CEBC [49] is the closest published method that edits unsupported object mentions in VLM output with an external detector: a detector-score threshold is set as a quantile of the scores of hallucinated mentions in base captions, a VLM rewrites unsupported mentions, and a final filter deletes what remains. It states no theorem, assumes no sampling model and does not place the revised text under a calibrated guarantee. §7 compares the three in detail.

**Shift caused by the predictor.** Editing a generation changes what follows. This is feedback covariate shift [18], to which weighted conformal prediction [59] does not apply; online adaptation

[23, 37] and reweighting [52] address parts of it. §8 shows that in generative correction the obstacle is stronger than loss of exchangeability.

**What is new here.** A selective predictor whose action space includes substitution and whose certificate covers the emitted mixture; a protocol in which the proved procedure and the evaluated procedure coincide; and measurements that bound what the mechanism contributes: attribution against score quality, accounting of what a repair changes, the fragility of the fixed sequence at small calibration sizes, and the failure of mid-sequence repair.

### 3 Problem setup

#### 3.1 Units, risk and the guarantee

Let  $Z = (X, A^*)$  be a query–answer pair, where  $X$  contains an image and a question and  $A^* \in \{0, 1\}$  is the gold answer. A VLM returns  $\hat{a}(X)$ . A second predictor (channel 2) returns an answer  $b(X)$  and an evidence margin  $m(X) \geq 0$ . Write  $Y = \mathbf{1}\{\hat{a}(X) = A^*\}$  and  $Y^{(2)} = \mathbf{1}\{b(X) = A^*\}$  for the two correctness indicators and  $\mu = \mathbb{P}[Y = 0]$  for the base risk.  $\alpha$  is always a risk target and  $\delta$  the failure probability of the certificate.

A *policy*  $\pi$  maps  $X$  to one of three actions: emit  $\hat{a}(X)$ , emit  $b(X)$ , or abstain. Let  $\mathcal{E}_\pi(X) = \mathbf{1}\{\pi \text{ emits}\}$  be the emission indicator and  $W_\pi = \mathcal{E}_\pi \cdot \mathbf{1}\{\text{emitted answer} \neq A^*\}$  the indicator of an emitted error. The *risk* and *coverage* of  $\pi$  are

$$\mathcal{R}(\pi) = \frac{\mathbb{E}[W_\pi]}{\mathbb{P}[\mathcal{E}_\pi = 1]}, \quad \mathcal{C}(\pi) = \mathbb{P}[\mathcal{E}_\pi = 1],$$

with the convention  $\mathcal{R}(\pi) = 0$  when  $\mathbb{P}[\mathcal{E}_\pi = 1] = 0$ . The same convention applies to a realised abort: a procedure that certifies nothing emits nothing, has risk 0 and coverage 0, and every coverage figure in this paper includes aborted splits as zero coverage.

**Assumption 1** (Sampling unit). The calibration units  $Z_1, \dots, Z_n$  are i.i.d. draws from a distribution  $P$ , independent of the data used to build the policy family (the *fit* data), and the unit is the item.

Exchangeability is not enough here. Under exchangeability alone, sampling without replacement from a finite pool makes the number of emitted errors hypergeometric rather than binomial, and the exact binomial bound used below is not valid. When several questions share an image the item is not an i.i.d. unit; §4.4 states what remains true and measures what does not.

**Assumption 2** (Pre-specified policy family). The candidate policies form a finite, ordered family  $\pi_1, \dots, \pi_M$  whose members and order are functions of the fit data and of constants only, never of calibration outcomes or calibration features.

Calibration is a statement about the *population* risk  $\mathcal{R}(\pi_j)$  of the policy the calibration sample selects; it is not a statement about any finite test fold, whose realised error rate carries binomial noise of its own.

### 3.2 Filtering and the floor it cannot cross

A *filtering* policy uses a score  $s(X) \in [0, 1]$ , large when the model is likely right, and emits  $\hat{a}(X)$  when  $s(X) \geq t$ .

**Proposition 1** (Abstention floor; Kotte [33]). *Let  $\mu > \alpha$ . Any selective predictor whose emitted answer is always  $\hat{a}(X)$  and whose risk is at most  $\alpha$  satisfies*

$$1 - \mathcal{C} \geq \frac{\mu - \alpha}{1 - \alpha}.$$

*Proof.* Split the error mass by emission,  $\mu = \mathcal{RC} + \mathbb{P}[Y = 0 \mid \text{abstain}](1 - \mathcal{C}) \leq \alpha\mathcal{C} + (1 - \mathcal{C})$ . The inequality uses that abstaining does not change  $Y$ . Solving for  $\mathcal{C}$  gives  $\mathcal{C} \leq (1 - \mu)/(1 - \alpha)$ .  $\square$

The proof uses one premise: an abstained item’s model answer is still wrong. It is a statement about the action space. Because  $\alpha$  is chosen by the user and  $\mu$  is estimated from data, the floor can be evaluated before calibration only up to the uncertainty of  $\hat{\mu}$ ; a feasibility check should use a confidence bound on  $\mu$ , not its plug-in value.

### 3.3 Two channels

**Definition 1** (Channels). *Channel 1* is a correctness score  $s(X) \approx \mathbb{P}[Y = 1 \mid \phi(X)]$  fitted on features of the model’s own output together with grounding features. *Channel 2* is a distinct predictor  $b(X)$  that answers the same question without using the VLM’s answer, together with a margin  $m(X)$  measuring how decisively it does so.

We call channel 2 a *distinct evidence source*. Statistical independence between  $Y$  and  $Y^{(2)}$  is neither assumed nor needed for validity: Theorem 1 holds for any fixed policy and needs only the sampling model. What distinctness buys is usefulness. A repair region helps only if  $\mathbb{P}[Y^{(2)} = 1 \mid \text{region}]$  is high, which requires channel 2’s errors to differ from the model’s inside that region (§6.6 shows what happens when channel 2 is the negation of the model’s answer). Our instantiation also violates independence in the other direction: the detector score is an input to channel 1.

**Instantiation.** Channel 2 is OWLv2 [48] queried with the object named in the question. With detection confidence  $o(X)$  and a fixed operating point  $\theta = 0.15$ ,  $b(X) = \mathbf{1}\{o(X) \geq \theta\}$  and  $m(X) = |o(X) - \theta|$ . A confident negative reaches only  $m \leq \theta$  while positives reach  $1 - \theta$ , so with a single gate on  $m$  every repair says “the object is present”. Section 6.3 evaluates a polarity-symmetric gate as well. When no detector value exists (the object phrase could not be parsed, or the question concerns an attribute or relation), the item keeps channel 1’s decision, receives  $m = -1$ , and can never be repaired.

Table 1: Notation.

|                           |                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------|
| $\alpha, \delta$          | risk target; failure probability of the certificate                                               |
| $\mu$                     | base risk $\mathbb{P}[Y = 0]$                                                                     |
| $Y, Y^{(2)}$              | correctness indicators of the model and of channel 2                                              |
| $s(X), m(X)$              | channel-1 score; channel-2 evidence margin                                                        |
| $t_1 > \dots > t_M$       | acceptance thresholds, score <i>values</i> fixed from fit data                                    |
| $g$                       | repair-gate value on $m$ , fixed from fit or external data                                        |
| $\mathcal{E}_j, K_j, V_j$ | emission indicator, emitted count and emitted-error count of policy $j$ on the calibration sample |
| $U(e, k; \delta)$         | one-sided Clopper–Pearson upper bound                                                             |
| $k_{\text{start}}(e_0)$   | smallest $k$ with $U(e_0, k; \delta) \leq \alpha$                                                 |

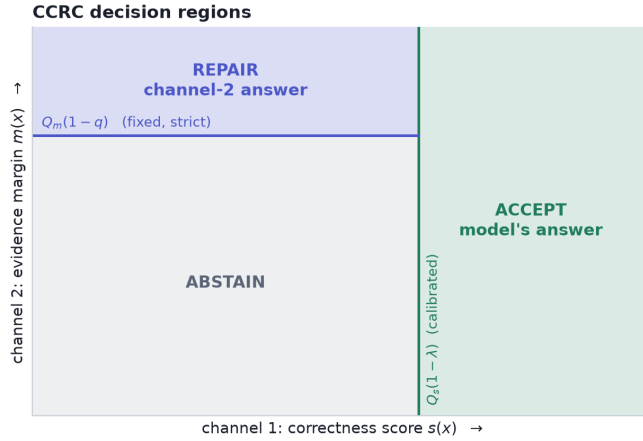


Figure 1: The  $(s, m)$  plane under policy (1). The acceptance threshold moves along the sequence; the repair gate is fixed, so the accepted region grows while the repair region shrinks.

## 4 Certified-correction risk control

### 4.1 The policy family

Fix a gate value  $g$  and a descending grid of acceptance thresholds  $t_1 > \dots > t_M$  on the channel-1 score. Policy  $\pi_j$  acts as

$$\pi_j(x) = \begin{cases} \text{ACCEPT } \hat{a}(x), & s(x) \geq t_j, \\ \text{REPAIR WITH } b(x), & s(x) < t_j \text{ and } m(x) \geq g, \\ \text{ABSTAIN}, & \text{otherwise.} \end{cases} \quad (1)$$

Figure 1 shows the induced partition of the  $(s, m)$  plane. The emitted set of  $\pi_j$  is the accepted set together with the repaired set, and an emitted error is a wrong accepted answer or a wrong channel-2 answer: with  $A_j = \{s \geq t_j\}$  and  $R_j = \{s < t_j, m \geq g\}$ ,

$$W_j = \mathbf{1}\{X \in A_j\}(1 - Y) + \mathbf{1}\{X \in R_j\}(1 - Y^{(2)}). \quad (2)$$

Because  $g$  does not depend on  $j$ , the emitted set  $A_j \cup \{m \geq g\}$  grows as  $t_j$  falls. Filtering is the case  $g = \infty$ .

## 4.2 How thresholds and gate are fixed

Both are values, and both are computed from the fit fold, which is disjoint from the calibration fold and from the test fold.

- *Score grid.* The channel-1 model is fitted on the fit fold. Its out-of-fold scores on that fold (image-grouped five-fold cross-fitting inside the fit fold) define the grid  $t_j = Q_{1-\lambda_j}$  of their empirical distribution, with  $\lambda_1 < \dots < \lambda_{40} = 1$  equally spaced. Using out-of-fold scores keeps the quantile-to-value map faithful to out-of-sample behaviour; an in-sample fit is mildly overconfident on its own data.
- *Start.* The first hypothesis is where  $c k_{\text{start}}(e_0)$  items are expected to be emitted,  $\lambda_1 = \min\{0.9, \max\{0.05, c k_{\text{start}}(e_0)/n_{\text{cal}}\}\}$ , with  $k_{\text{start}}(e_0) = \min\{k : U(e_0, k; \delta) \leq \alpha\}$  the smallest emitted count at which  $e_0$  observed errors still certify the target (Lemma 1). The default is  $e_0 = 1$ ,  $c = 1.5$ : start where one error can be tolerated, with half as much again as headroom for fold-to-fold variation in  $K$ . The start is computed from  $\alpha$ ,  $\delta$  and  $n_{\text{cal}}$  alone, before any label is read; §6.4 ablates it.
- *Repair gate.*  $g = Q_{1-q}(m)$  over grounded fit-fold items. Three ways of fixing  $q$  keep the guarantee: (a) *external*:  $q$  is chosen once on a different benchmark by running the same procedure there and keeping the candidate (filtering, or  $q \in \{0.05, 0.10, 0.25, 0.50\}$ ) that certifies the most coverage; (b) *fit-fold*: the same choice is made inside each split on the fit fold alone; (c) *union*: three gates  $\{0.05, 0.10, 0.25\}$  are each certified at  $\delta/3$  and the certified policy with the largest calibration coverage is emitted.

## 4.3 Validity

**Theorem 1** (CCRC validity). *Under Assumptions 1 and 2, let  $\hat{j}$  be the last index passed by the fixed-sequence test that proceeds through  $j = 1, 2, \dots$  while  $U(V_j, K_j; \delta) \leq \alpha$  on the calibration sample and stops at the first failure ( $\hat{j} = 0$  if the first test fails, in which case nothing is emitted). Then*

$$\mathbb{P}[\mathcal{R}(\pi_{\hat{j}}) \leq \alpha] \geq 1 - \delta,$$

where the probability is over the calibration sample, conditional on the fit data.

*Proof.* Condition on the fit data, so the policies are fixed. For policy  $j$  let  $K_j = \sum_i \mathcal{E}_j(X_i)$  and  $V_j = \sum_i W_j(Z_i)$ , and consider  $H_j : \mathcal{R}(\pi_j) > \alpha$ . Emission is a function of  $X$  alone. By Assumption 1 the units are independent, so conditional on the set of emitted units (equivalently, on the emission pattern), the emitted units are i.i.d. draws from  $P(\cdot \mid \mathcal{E}_j = 1)$  and each has error indicator  $W_j \sim \text{Bernoulli}(\mathcal{R}(\pi_j))$ , independently across units. Hence  $V_j \mid \{K_j = k\} \sim \text{Bin}(k, \mathcal{R}(\pi_j))$ , and by construction of the Clopper–Pearson bound,  $\mathbb{P}[U(V_j, K_j; \delta) \leq \alpha \mid K_j = k] \leq \delta$  whenever  $\mathcal{R}(\pi_j) > \alpha$  and  $k \geq 1$ ; for  $k = 0$  the bound is  $1 > \alpha$ . So  $\phi_j = \mathbf{1}\{U \leq \alpha\}$  is a level- $\delta$  test of  $H_j$ .

Let  $j_0 = \min\{j : H_j \text{ is true}\}$ , if it exists. The sequence reaches an index  $\geq j_0$  only if  $H_{j_0}$  is rejected, an event of probability at most  $\delta$ . On the complement,  $\hat{j} < j_0$ , so every index the procedure can return has  $\mathcal{R}(\pi_{\hat{j}}) \leq \alpha$ . If no  $H_j$  is true the claim is trivial, and  $\hat{j} = 0$  emits nothing and has risk 0 by convention.  $\square$

The proof uses neither monotonicity of the risk along the sequence nor nesting of the emitted sets; nesting is what makes “the last index that passed” a coverage-maximising return. It also does not use whether  $W_j$  refers to the model’s answer or to channel 2’s, which is why Proposition 1 does not bind: its premise restricts the action space, and the certificate never refers to it.

**Corollary 1** (Gate fixed on other data, or chosen on the fit fold). *If  $g$  is a constant, is computed from external data independent of the calibration sample, or is computed from the fit data (including by a selection rule applied to the fit data alone), the policy family satisfies Assumption 2 and Theorem 1 holds at the full level  $\delta$ .*

**Corollary 2** (Union over a pre-specified set of gates). *Let  $\mathcal{Q}$  be a finite set of gate values fixed in advance. Run the fixed-sequence test for each at level  $\delta/|\mathcal{Q}|$  and emit any certified policy, chosen by any rule that may use the calibration sample. Then the emitted policy has risk at most  $\alpha$  with probability at least  $1 - \delta$ .*

*Proof.* By Theorem 1 each family fails with probability at most  $\delta/|\mathcal{Q}|$ ; on the complement of the union of these events every certified policy is compliant.  $\square$

A gate chosen by looking at the calibration or test outcomes of the benchmark under evaluation is not covered by either corollary. Section 6.1 reports the  $q = 0.10$  gate that earlier experiments selected for that reason as an *uncertified* reference arm.

**Lemma 1** (Minimal certifiable emitted set).  *$U(0, k; \delta) = 1 - \delta^{1/k}$ , so  $e_0 = 0$  errors certify  $\alpha$  iff  $k \geq k_{\text{start}}(0) = \lceil \ln \delta / \ln(1 - \alpha) \rceil$ , which is 22 at  $\alpha = \delta = 0.10$ . For  $e_0 \geq 1$ ,  $k_{\text{start}}(e_0)$  is the smallest  $k$  with  $\mathbb{P}[\text{Bin}(k, \alpha) \leq e_0] \leq \delta$ ; at  $\alpha = \delta = 0.10$  it is 38, 52, 65 for  $e_0 = 1, 2, 3$ , and at  $\delta/3$  it is 33, 51, 66, 81 for  $e_0 = 0, \dots, 3$ .*

*Proof.*  $\mathbb{P}[\text{Bin}(k, p) \leq e_0]$  is decreasing in  $p$  and equals  $\delta$  at  $p = U(e_0, k; \delta)$ , so  $U(e_0, k; \delta) \leq \alpha$  iff  $\mathbb{P}[\text{Bin}(k, \alpha) \leq e_0] \leq \delta$ . For  $e_0 = 0$  this is  $(1 - \alpha)^k \leq \delta$ .  $\square$

Starting the sequence at  $e_0 = 0$  makes the first test fail if a single error falls in the first emitted set, and because the sequence stops at the first failure, that is an abort. An analytic start at  $e_0 \geq 1$  removes this sensitivity to one error but places the first hypothesis at a more permissive threshold, which a weak score may fail for a different reason. Neither is better in general, which is why §6.4 reports the whole range.

#### 4.4 Clustered units

When an image carries several questions, the item is not an i.i.d. unit; the image is. Two exact options exist. If calibration uses one item drawn uniformly at random from each of  $n$  i.i.d. images, the units are i.i.d. and Theorem 1 applies to the risk under that sampling scheme, which equals the item-level risk when images have equal numbers of items. If calibration uses all items, positive within-image correlation of errors makes  $V_j | K_j$  over-dispersed relative to the binomial and the bound is anti-conservative by an amount that grows with the intra-image correlation; §6.8 measures this in a known-population simulation and §5 measures the correlation in the data we use.

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**Algorithm 1** CCRC. Folds  $\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{cal}}$  are disjoint and unions of images;  $\alpha, \delta, e_0, c$  are fixed in advance.

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1: fit  $s$  on  $\mathcal{D}_{\text{fit}}$ ; compute out-of-fold scores  $\tilde{s}$  on  $\mathcal{D}_{\text{fit}}$ 
2: choose gate fraction  $q$  (external, fit-fold, or union);  $g \leftarrow Q_{1-q}\{m_i : i \in \mathcal{D}_{\text{fit}}\}$ 
3:  $\lambda_1 \leftarrow \min\{0.9, \max\{0.05, c k_{\text{start}}(e_0)/|\mathcal{D}_{\text{cal}}|\}\}$ ;  $t_j \leftarrow Q_{1-\lambda_j}(\tilde{s})$ ,  $j = 1..M$ 
4:  $\hat{j} \leftarrow 0$ 
5: for  $j = 1$  to  $M$  do
6:    $A \leftarrow \{i \in \mathcal{D}_{\text{cal}} : s_i \geq t_j\}$ ,  $R \leftarrow \{i \notin A : m_i \geq g\}$ ;  $K \leftarrow |A| + |R|$ 
7:    $V \leftarrow \sum_A (1 - Y_i) + \sum_R (1 - Y_i^{(2)})$ 
8:   if  $K > 0$  and  $U(V, K; \delta) \leq \alpha$  then  $\hat{j} \leftarrow j$ 
9:   else break
10:  end if
11: end for
12: deploy: if  $\hat{j} = 0$  abstain everywhere; else accept if  $s(x) \geq t_{\hat{j}}$ , repair if  $m(x) \geq g$ , abstain otherwise

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## 4.5 The rank-indexed variant

An implementer who has no pre-specified grid takes thresholds as quantiles of the calibration sample itself,  $t_j = Q_{1-\lambda_j}(s_{\text{cal}})$ , and a repair gate  $g = Q_{1-q}(m_{\text{cal}})$ . Selecting the top  $\lceil \lambda n \rceil$  items by score makes  $K$  deterministic and makes the emitted set a function of all calibration features jointly; then  $V$  is Poisson-binomial rather than binomial and the argument above does not apply. We treat this variant as the conventional baseline and quantify its behaviour in §6.2 and §6.8, without claiming a theorem for it.

## 5 Experimental protocol

**Models, benchmarks and signals.** Backbones are LLaVA-1.5-7B [44] in 4-bit, Qwen2-VL-2B-Instruct [64], and LLaVA decoded with visual contrastive decoding [39]. Channel 2 is OWLv2-base [48] with  $\theta = 0.15$ . Benchmarks are the POPE adversarial split [40] and the AMBER discriminative benchmark [63]; MME, GQA and HallusionBench appear only where the detector does not apply (§6.7). Per-item signals were extracted once and are released, so every analysis below runs on a CPU. We did not rerun the VLMs.

**Cohorts and the unit of splitting.** The extraction streamed the first  $N$  rows of each public benchmark. We recovered the image identifier of every item from the benchmark metadata and verified the reconstruction against the cached gold labels, object phrases and categories (all agree). Table 2 reports what enters each cell. The POPE items are six questions per image (three present, three absent objects), so there are 75, 100 and 250 distinct images in the three POPE cell sizes, and the smaller POPE cells are subsets of the largest one’s images. AMBER contributes 1.3 items per image. Folds are unions of whole images, drawn at random with a third of the images per fold. The intra-image correlation of the model’s errors is slightly *negative* ( $-0.04$  to  $-0.09$ ): the balanced yes/no design makes an error on one question of an image mildly predictive of a correct answer on another.

Table 2: Cohort flow. “Detector value” is the number of items for which the detector returned a score; the remaining items stay in every analysis, keep the channel-1 decision, and cannot be repaired. In POPE the exclusions are object phrases the extraction script failed to parse (POPE contains the typo “imange”); in AMBER they are the attribute and relation queries, which name no object. ICC is the intra-image correlation of the model’s errors.

| Setting             | items | images | items/img | ICC   | detector value | excluded from the grounded cohort | images/fold | $\mu$ (%) |
|---------------------|-------|--------|-----------|-------|----------------|-----------------------------------|-------------|-----------|
| POPE-1500, LLaVA    | 1500  | 250    | 6.0       | −0.04 | 1500           | 0 unparsed object phrases         | 83/83/84    | 18.1      |
| POPE-adv, LLaVA     | 450   | 75     | 6.0       | −0.05 | 444            | 6 unparsed object phrases         | 25/25/25    | 17.3      |
| POPE-adv, Qwen2-VL  | 600   | 100    | 6.0       | −0.04 | 591            | 9 unparsed object phrases         | 33/33/34    | 12.7      |
| POPE-adv, LLaVA+VCD | 600   | 100    | 6.0       | −0.04 | 591            | 9 unparsed object phrases         | 33/33/34    | 19.7      |
| AMBER(d), LLaVA     | 500   | 392    | 1.3       | −0.09 | 228            | 272 attribute/relation queries    | 130/130/132 | 21.4      |

**Primary analysis and missing evidence.** The primary cohort contains all items. Items with no detector value receive zeroed grounding features and a missing-evidence flag in the score, and a margin  $m = -1$  so they are never repaired. A secondary analysis restricted to items with a detector value appears in Appendix A.3. Excluding the items would be an analysis of the detector’s coverage, not of deployment: a system must decide what to do with a question the detector cannot parse.

**Procedure.** Each split draws three image-disjoint folds (fit, calibration, test); the score is a logistic regression with five features (the VLM’s probability, its confidence margin, the detector score, the detector score signed by the VLM’s answer, and the missing flag); thresholds and gate are fixed from the fit fold as in §4.2; the fixed-sequence test uses Clopper–Pearson bounds at  $\delta = 0.10$ ; and the deployed policy is evaluated on the test fold. We use 400 paired splits per cell, so every arm in a comparison sees identical folds.

**Uncertainty.** Repeated random splits of one benchmark measure how sensitive the procedure is to the split. They do not measure how the result would change on a fresh benchmark: the interval over splits shrinks like  $1/\sqrt{\text{splits}}$  while uncertainty about the dataset does not shrink at all. We therefore report gains as the mean over the 400 splits with a 95% percentile interval from an *outer image-level bootstrap*: 300 bootstrap datasets are drawn by resampling images with replacement, the full protocol is run on each with 20 splits, and the interval is taken over bootstrap datasets. A duplicated image keeps its identifier and is always placed in one fold. Because a bootstrap dataset contains only about 63% distinct images, the interval is somewhat wider than the one a fresh sample of the same size would give. We do not report  $p$ -values over splits.

**Validity audits.** Realised test-fold risk carries binomial noise and cannot audit a population guarantee: a policy whose true risk is 0.08 exceeds 0.10 on a 150-item test fold with substantial probability. We report two statistics per cell. *exc* is the fraction of non-aborted splits whose test-fold risk exceeds  $\alpha$ ; under validity it is bounded by  $\delta$  plus finite-fold noise. *cv* is the fraction of splits for which the one-sided 90% Clopper–Pearson *lower* bound on the test-fold risk exceeds  $\alpha$  (a confident violation); if the population risk is at most  $\alpha$  it is at most 0.1 per split by the same argument as the theorem. Test folds are used for nothing else: no score, threshold, gate or start is selected on them. An exact audit requires a known population, which the simulation in

Table 3: Certified coverage (%), with the abort rate (%) in parentheses,  $\delta = 0.10$ , 400 paired splits, all-item cohort. Gain is the mean paired difference over splits against filtering; the interval is the 95% percentile interval from the outer image-level bootstrap (§5), and  $P(\text{gain} > 0)$  is the fraction of bootstrap datasets with a positive gain.

| Setting             | $\alpha$ | Filter       |              | CCRC, external gate |                      |           | CCRC, fit-fold gate |               |
|---------------------|----------|--------------|--------------|---------------------|----------------------|-----------|---------------------|---------------|
|                     |          | cov. (abort) | cov. (abort) | gain [95% CI]       | $P(\text{gain} > 0)$ |           | cov. (abort)        | gain [95% CI] |
| POPE-1500, LLaVA    | 0.10     | 68.2 (0)     | 71.1 (0)     | +2.9 [+1.4, +11.8]  | 0.99                 | 77.0 (1)  | +8.8 [−8.0, +18.8]  |               |
|                     | 0.15     | 85.5 (0)     | 91.3 (0)     | +5.8 [+1.3, +9.3]   | 0.98                 | 92.6 (0)  | +7.1 [+3.6, +10.7]  |               |
| POPE-adv, LLaVA     | 0.10     | 44.5 (25)    | 48.2 (22)    | +3.7 [−2.3, +10.3]  | 0.91                 | 41.4 (34) | −3.1 [−13.8, +9.3]  |               |
|                     | 0.15     | 72.5 (2)     | 80.5 (2)     | +7.9 [−5.9, +29.0]  | 0.92                 | 82.0 (2)  | +9.4 [−3.3, +28.8]  |               |
| POPE-adv, Qwen2-VL  | 0.10     | 76.9 (1)     | 79.4 (0)     | +2.6 [−1.5, +9.5]   | 0.95                 | 71.2 (10) | −5.7 [−25.1, +10.5] |               |
|                     | 0.15     | 94.6 (0)     | 95.4 (0)     | +0.8 [−13.1, +10.7] | 0.70                 | 95.0 (0)  | +0.4 [−7.5, +8.9]   |               |
| POPE-adv, LLaVA+VCD | 0.10     | 45.8 (14)    | 47.8 (11)    | +2.0 [−1.9, +7.2]   | 0.91                 | 39.6 (30) | −6.2 [−13.4, +8.9]  |               |
|                     | 0.15     | 74.4 (0)     | 76.0 (1)     | +1.6 [−10.4, +20.5] | 0.76                 | 76.1 (0)  | +1.7 [−10.3, +17.6] |               |
| AMBER(d), LLaVA     | 0.10     | 55.2 (3)     | 56.3 (6)     | +1.1 [−15.1, +11.5] | 0.65                 | 55.8 (6)  | +0.6 [−8.5, +9.1]   |               |
|                     | 0.15     | 74.1 (0)     | 76.1 (0)     | +1.9 [−10.6, +7.9]  | 0.73                 | 75.3 (0)  | +1.2 [−8.0, +5.9]   |               |

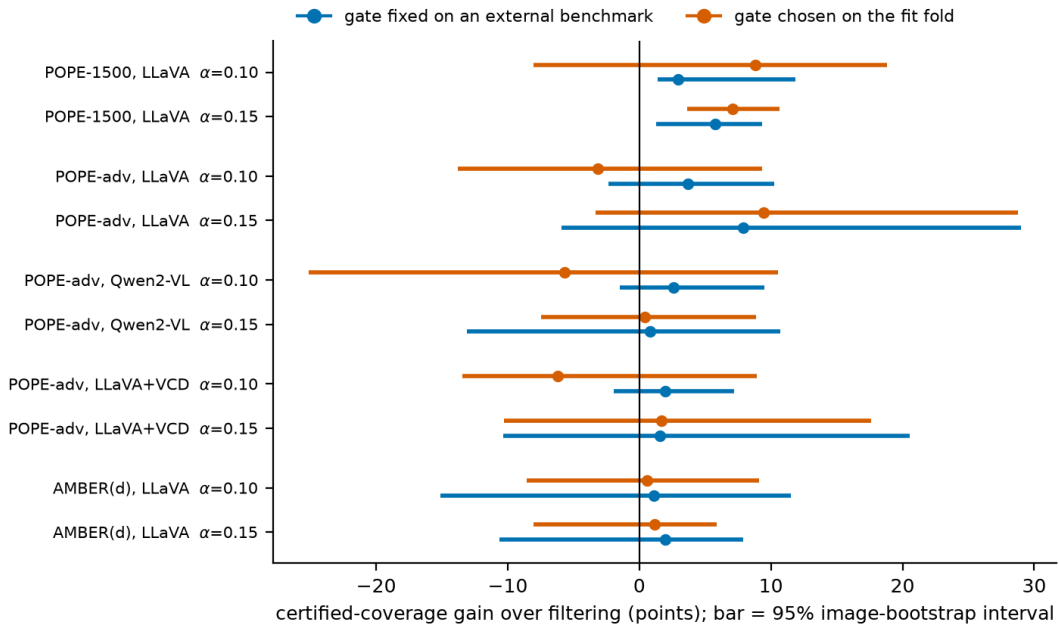


Figure 2: Gain in certified coverage over filtering. Points are means over 400 splits of the original benchmark; bars are 95% image-bootstrap intervals.

§6.8 provides.

## 6 Results

### 6.1 Certified coverage

Table 3 compares CCRC with filtering on identical folds at  $\alpha \in \{0.10, 0.15\}$ ,  $\delta = 0.10$ , all items, and Figure 2 plots the gains with their bootstrap intervals. Two certified variants appear. In the *external gate* variant  $q$  is a constant chosen once on a different benchmark (AMBER for the POPE cells, POPE-1500 for AMBER; the choice is  $q = 0.10$  at  $\alpha = 0.10$  and 0.25 at 0.15 for the POPE cells, and 0.50 for AMBER). In the *fit-fold gate* variant  $q$  is chosen inside each split on the fit fold, with filtering as one candidate.

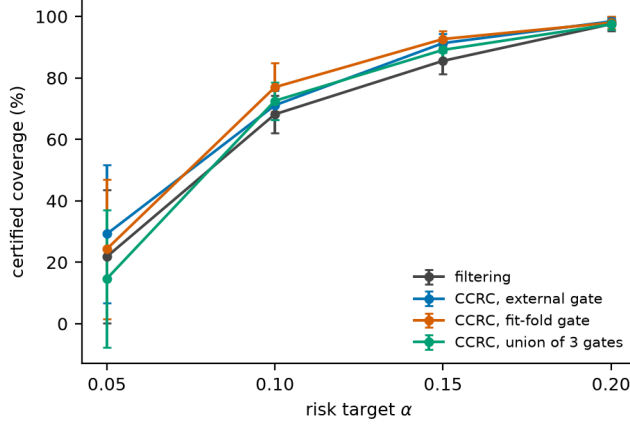


Figure 3: Certified coverage against the risk target on POPE-1500. Bars are one standard deviation over splits.

With the external gate the gain is positive in all ten cells, from +0.8 to +7.9 points, and the share of bootstrap datasets with a positive gain is 0.65–0.99. The interval excludes zero on POPE-1500 at both targets (+2.9 at  $\alpha = 0.10$ , +5.8 at 0.15) and on no other cell. The other cells rest on 75–392 images, folds of 25–130 images, and filtering aborts on 25% and 14% of splits on the two smallest; the data support the sign more than the size. The same data do not support a larger claim than that, and the per-split standard deviations (Appendix A.1) are 2–15 points.

The fit-fold gate is the more flexible variant and the more variable. On POPE-1500 it selects  $q = 0.5$  in 84–92% of splits and gains +8.8 and +7.1 points, a larger gain than any fixed gate; on the four cells with 150–200 calibration items its gain ranges from  $-6.2$  to +9.4 points and is negative in three of the four small cells at  $\alpha = 0.10$ , because a gate chosen on a fit fold of 150 items is itself noisy and a wrong choice costs more than the right one returns. The union over three gates avoids selection noise and pays  $\delta/3$  per gate: it gains +4.4 and +3.6 points on POPE-1500 and loses up to 16.5 points on the cells with fewer than 200 calibration items (Appendix A.1). The  $q = 0.10$  gate that earlier experiments selected after looking at these benchmarks gains +0.6 to +3.9 points in every cell. We report it as an uncertified reference and do not use it for any claim, since a gate chosen with knowledge of the benchmarks is exactly what Theorem 1 does not cover.

Coverage rises with  $\alpha$  for every arm (Figure 3); the gap between CCRC and filtering is largest at moderate targets. The audit columns of Appendix Table 16 are discussed in §6.8.

## 6.2 What each protocol change does

Table 4 introduces the changes one at a time, from a conventional implementation (item-level folds, thresholds and gate as quantiles of the calibration fold, a start at  $k_{\text{start}}(0)$  and a fixed  $q = 0.10$ ) to the protocol of this paper; the  $\alpha = 0.15$  version is in Appendix A.2.

Image-grouped folds change coverage by at most 1.2 points and lower the test-fold exceedance on the four POPE cells from 0.14, 0.11, 0.16, 0.12 to 0.10, 0.09, 0.09, 0.12. Replacing calibration quantiles by fit-fold values costs coverage when the fixed sequence starts at  $e_0 = 0$ , because the first hypothesis then lands at a threshold where a single error aborts the test, and the abort rate

Table 4: Filtering / CCRC certified coverage (%) at  $\alpha = 0.10$  as the protocol changes. Rungs L0–L3 use the fixed gate  $q = 0.10$ , rung L4 chooses the gate on the fit fold, and rung L5 keeps items with no detector value. L0–L4 use the grounded cohort of each cell; L5 uses all items, so its absolute level for AMBER is lower: the attribute and relation queries are harder and cannot be repaired.

|                                                                                      | POPE-1500, LLaVA | POPE-adv, LLaVA | POPE-adv, Qwen2-VL | POPE-adv, LLaVA+VCD | AMBER(d), LLaVA |
|--------------------------------------------------------------------------------------|------------------|-----------------|--------------------|---------------------|-----------------|
| conventional protocol (item folds, calibration quantiles, $e_0=0$ , fixed $q=0.10$ ) | 68.6/71.8        | 44.0/47.1       | 77.9/79.7          | 48.8/51.0           | 91.8/81.6       |
| + image-grouped folds                                                                | 68.2/71.3        | 43.1/48.3       | 77.6/79.5          | 47.9/50.8           | 92.0/81.9       |
| + fixed value grid and gate (fit fold)                                               | 67.5/70.9        | 35.0/42.3       | 75.5/77.7          | 39.9/45.3           | 90.9/85.1       |
| + analytic start $e_0=1$                                                             | 68.2/71.1        | 43.6/47.4       | 75.6/78.0          | 45.5/47.6           | 90.2/90.0       |
| + gate chosen on the fit fold                                                        | 68.2/77.0        | 43.6/39.4       | 75.6/68.3          | 45.5/39.7           | 90.2/89.8       |
| + all-item cohort                                                                    | 68.2/77.0        | 44.5/41.4       | 76.9/71.2          | 45.8/39.6           | 55.2/55.8       |

Table 5: Accounting of repaired items, external-gate variant, all items, pooled over test folds of 400 splits. “Repaired” and “answer changed” are percentages of test items; the next five columns are percentages of repaired items. “FP corrected” is the number of the model’s false-positive errors (it said yes, gold is no) whose emitted answer was corrected, over the number of such errors; “FN corrected” is the share of the model’s false-negative errors corrected.

| Setting             | $\alpha$ | $q$  | % of test items |                | % of repaired items |             |             |              |              | FP corrected | FN corrected (%) |
|---------------------|----------|------|-----------------|----------------|---------------------|-------------|-------------|--------------|--------------|--------------|------------------|
|                     |          |      | repaired        | answer changed | changed             | wrong→right | right→wrong | agree, right | agree, wrong |              |                  |
| POPE-1500, LLaVA    | 0.10     | 0.1  | 1.88            | 1.11           | 59                  | 59          | 0.3         | 41           | 0.0          | 0/11482      | 8.9              |
|                     | 0.15     | 0.25 | 4.81            | 3.33           | 69                  | 62          | 7.3         | 30           | 0.8          | 0/11482      | 24.0             |
| POPE-adv, LLaVA     | 0.10     | 0.1  | 1.93            | 0.69           | 36                  | 34          | 1.6         | 64           | 0.0          | 0/3019       | 6.1              |
|                     | 0.15     | 0.25 | 5.29            | 2.19           | 41                  | 39          | 2.2         | 57           | 1.3          | 0/3815       | 19.0             |
| POPE-adv, Qwen2-VL  | 0.10     | 0.1  | 0.90            | 0.04           | 4                   | 4           | 0.0         | 96           | 0.0          | 0/2724       | 0.4              |
|                     | 0.15     | 0.25 | 1.11            | 0.39           | 35                  | 27          | 8.6         | 56           | 8.8          | 0/2724       | 3.2              |
| POPE-adv, LLaVA+VCD | 0.10     | 0.1  | 0.68            | 0.39           | 57                  | 57          | 0.4         | 43           | 0.0          | 0/9337       | 5.8              |
|                     | 0.15     | 0.25 | 2.68            | 1.18           | 44                  | 35          | 9.4         | 51           | 4.6          | 0/10431      | 13.9             |
| AMBER(d), LLaVA     | 0.10     | 0.5  | 3.64            | 1.67           | 46                  | 45          | 1.4         | 48           | 5.6          | 115/4352     | 9.8              |
|                     | 0.15     | 0.5  | 1.76            | 1.08           | 61                  | 61          | 0.0         | 29           | 9.9          | 58/4624      | 6.8              |

of filtering on POPE-adv LLaVA rises from 19% to 34%; the analytic start at  $e_0 = 1$  recovers most of it. The gain of repair over filtering is +2.9 to +3.5 points on POPE-1500 under L0–L3, stable under all of these changes. On AMBER the loss of  $-10.2$  points under the conventional protocol is not a property of repair: it falls to  $-5.8$  with value-indexed thresholds and to  $-0.3$  with the analytic start, and is +0.6 on the all-item cohort.

### 6.3 What a repair is

The reviewer-style question for any correction method is how often it changes the model’s answer, and whether the changes are right. Table 5 separates the repaired items (those emitted through channel 2) into those where channel 2’s answer differs from the model’s and those where the two agree, and counts wrong-to-right and right-to-wrong changes.

The repaired mass is 0.7–3.6% of test items at  $\alpha = 0.10$ , and the answer is changed for 0.04–1.7% of test items. Among repaired items the detector agrees with the model on 41–96%: these items are *re-emitted with detector confirmation*, which is not correction and is a different benefit, namely that an item the correctness score would have abstained on is emitted because an independent source confirms it. Where the answer is changed it is almost always right (for example 2230 wrong-to-right against 13 right-to-wrong changes on POPE-1500). No false-positive error is corrected on the POPE cells (0 of 2,724–11,482) and 2.6% on AMBER with the larger gate  $q = 0.5$ ; between 0.4 and 9.8% of the model’s false-negative errors are corrected.

Table 6 repeats the accounting for one-sided and polarity-symmetric gates, each gating a fixed fraction  $q = 0.10$  of its own polarity’s margins. The symmetric gate admits negative-polarity repairs, which are highly precise, and these confirm the model’s own “no” answers; none changes

Table 6: Gate polarity at  $\alpha = 0.10$ ,  $q = 0.10$  per polarity. Gain over filtering in points, repaired items as a percentage of test items, and the number of false-positive errors corrected (pooled over 400 test folds). Pooled margin is the gate used elsewhere in the paper.

| Setting             | pooled margin (CCRC) |        |          | positive only |        |          | negative only |        |          | symmetric |        |          |
|---------------------|----------------------|--------|----------|---------------|--------|----------|---------------|--------|----------|-----------|--------|----------|
|                     | gain                 | rep. % | FP fixed | gain          | rep. % | FP fixed | gain          | rep. % | FP fixed | gain      | rep. % | FP fixed |
| POPE-1500, LLaVA    | +2.9                 | 1.88   | 0        | +1.3          | 0.80   | 0        | +0.2          | 0.07   | 0        | +1.4      | 0.87   | 0        |
| POPE-adv, LLaVA     | +3.7                 | 1.93   | 0        | +1.6          | 0.61   | 0        | +0.8          | 0.14   | 0        | +2.1      | 0.76   | 0        |
| POPE-adv, Qwen2-VL  | +2.6                 | 0.90   | 0        | +1.0          | 0.40   | 0        | +2.5          | 0.20   | 0        | +3.2      | 0.59   | 0        |
| POPE-adv, LLaVA+VCD | +2.0                 | 0.68   | 0        | +0.8          | 0.19   | 0        | +0.9          | 0.21   | 0        | +1.5      | 0.40   | 0        |
| AMBER(d), LLaVA     | +2.5                 | 2.84   | 0        | +1.1          | 0.61   | 0        | +0.1          | 0.03   | 0        | +1.1      | 0.64   | 0        |

Table 7: Gain decomposition (points of test coverage),  $q = 0.10$ , all items, conditional on both arms certifying.

| Setting             | $\alpha$ | threshold moves (%) | gain | (i) repairs at filter’s threshold | (ii) dilution [95% CI over splits] |
|---------------------|----------|---------------------|------|-----------------------------------|------------------------------------|
| POPE-1500, LLaVA    | 0.10     | 31                  | 2.79 | 1.89                              | 0.89 [0.74, 1.05]                  |
|                     | 0.15     | 25                  | 2.06 | 1.43                              | 0.63 [0.51, 0.75]                  |
| POPE-adv, LLaVA     | 0.10     | 24                  | 2.75 | 1.94                              | 0.81 [0.52, 1.10]                  |
|                     | 0.15     | 26                  | 3.04 | 1.59                              | 1.46 [0.99, 1.92]                  |
| POPE-adv, Qwen2-VL  | 0.10     | 18                  | 2.17 | 0.93                              | 1.24 [0.64, 1.84]                  |
|                     | 0.15     | 3                   | 0.37 | 0.31                              | 0.06 [0.01, 0.10]                  |
| POPE-adv, LLaVA+VCD | 0.10     | 12                  | 1.12 | 0.67                              | 0.45 [0.25, 0.66]                  |
|                     | 0.15     | 11                  | 1.14 | 0.49                              | 0.65 [0.28, 1.02]                  |
| AMBER(d), LLaVA     | 0.10     | 34                  | 3.61 | 2.84                              | 0.78 [0.33, 1.23]                  |
|                     | 0.15     | 27                  | 1.93 | 1.47                              | 0.46 [0.20, 0.72]                  |

an answer. The reason is structural in this instantiation: the model’s false positives are items where the detector also fires, so the detector’s strongly negative evidence occurs on items the model already got right.

**Where the gain comes from.** Repaired items are themselves emitted, so whenever the certified threshold does not move the gain over filtering equals the repaired mass at filtering’s own threshold. Table 7 splits the gain into this term and a dilution term, the extra accepted mass admitted because high-precision repairs lower the emitted risk and let the threshold open further; the gate is the continuity gate  $q = 0.10$ . The certified threshold differs between the two arms on 2.5–34% of splits, and the dilution term is 0.05 to 1.5 points; most of the gain is automatic.

## 6.4 The start of the sequence

Table 8 varies the number of errors the first hypothesis can tolerate,  $e_0 \in \{0, 1, 2, 3\}$  (so  $k_{\text{start}} = 22, 38, 52, 65$ ), and the placement multiplier  $c \in \{1.0, 1.5\}$ , for filtering and CCRC with the fit-fold gate. On POPE-1500 the choice is nearly irrelevant for CCRC (76.5–77.5%) and moves filtering by 2.6 points. On the smallest cell (150 calibration items) filtering ranges from 22% to 45% with abort rates from 24 to 67%, and CCRC follows. Neither extreme is good:  $e_0 = 0$  makes the first test fail on a single error; large  $e_0$  together with  $c = 1.5$  places the first hypothesis at a permissive threshold at which the score does not separate. We use  $e_0 = 1$ ,  $c = 1.5$  throughout, fixed before the final runs and following the analytic argument of Lemma 1; the table shows it is a reasonable region and that it was not selected on test outcomes. The practical consequence is that conclusions drawn on calibration folds of 150–200 items depend on the start

Table 8: Certified coverage (%) and abort rate (%) at  $\alpha = 0.10$  as the start of the fixed sequence varies, filtering and CCRC with the fit-fold gate (Appendix A.2 for  $\alpha = 0.05$  and  $0.15$ ).

| $c$ | $e_0$ | $k_{\text{start}}$ | POPE-1500, LLaVA |          | POPE-adv, LLaVA |           | POPE-adv, Qwen2-VL |           | POPE-adv, LLaVA+VCD |           | AMBER(d), LLaVA |           |
|-----|-------|--------------------|------------------|----------|-----------------|-----------|--------------------|-----------|---------------------|-----------|-----------------|-----------|
|     |       |                    | filter           | CCRC     | filter          | CCRC      | filter             | CCRC      | filter              | CCRC      | filter          | CCRC      |
| 1.0 | 0     | 22                 | 65.8 (4)         | 76.5 (1) | 22.2 (54)       | 33.8 (45) | 72.6 (5)           | 64.6 (19) | 22.9 (50)           | 29.9 (44) | 55.8 (0)        | 47.5 (20) |
| 1.0 | 1     | 38                 | 68.3 (0)         | 76.6 (1) | 41.2 (24)       | 40.7 (32) | 76.5 (0)           | 68.2 (13) | 42.8 (17)           | 39.1 (30) | 55.8 (0)        | 51.7 (12) |
| 1.0 | 2     | 52                 | 68.4 (0)         | 76.9 (1) | 44.4 (24)       | 41.8 (33) | 76.8 (0)           | 71.5 (10) | 45.2 (15)           | 39.0 (30) | 55.0 (2)        | 54.3 (8)  |
| 1.0 | 3     | 65                 | 68.2 (0)         | 77.2 (0) | 44.0 (28)       | 39.3 (40) | 76.0 (3)           | 69.5 (13) | 45.3 (16)           | 39.5 (31) | 54.3 (6)        | 54.9 (10) |
| 1.5 | 0     | 22                 | 67.5 (0)         | 76.8 (1) | 36.9 (30)       | 38.4 (36) | 76.5 (0)           | 68.6 (14) | 39.5 (22)           | 37.9 (32) | 55.7 (0)        | 49.7 (15) |
| 1.5 | 1     | 38                 | 68.2 (0)         | 77.0 (1) | 44.5 (25)       | 41.4 (34) | 76.9 (1)           | 71.2 (10) | 45.8 (14)           | 39.6 (30) | 55.2 (3)        | 55.8 (6)  |
| 1.5 | 2     | 52                 | 68.4 (0)         | 77.1 (1) | 38.5 (39)       | 36.2 (45) | 77.1 (4)           | 70.0 (14) | 43.8 (20)           | 37.6 (35) | 52.6 (11)       | 53.0 (14) |
| 1.5 | 3     | 65                 | 68.3 (0)         | 77.5 (0) | 22.4 (67)       | 22.2 (68) | 78.6 (2)           | 73.7 (10) | 33.0 (45)           | 29.4 (52) | 29.6 (54)       | 30.8 (52) |

Table 9: POPE-1500, filtering only: certified coverage (%) with abort rate in parentheses, and area under the risk–coverage curve at  $\alpha = 0.10$  (lower is better).

| Score                                        | $\alpha=0.05$ | $\alpha=0.10$ | $\alpha=0.15$ | AURC   |
|----------------------------------------------|---------------|---------------|---------------|--------|
| image–text similarity only (ConfLVLM scorer) | 0.0 (100)     | 0.5 (94)      | 3.4 (72)      | 0.1621 |
| VLM confidence only                          | 2.4 (93)      | 41.0 (18)     | 75.1 (5)      | 0.0768 |
| learned, VLM signals                         | 10.6 (61)     | 58.8 (1)      | 80.7 (0)      | 0.0678 |
| + image–text similarity                      | 9.0 (67)      | 56.8 (2)      | 80.5 (0)      | 0.0691 |
| + detection grounding                        | 21.8 (44)     | 68.2 (0)      | 85.5 (0)      | 0.0562 |
| + both                                       | 18.8 (46)     | 67.6 (0)      | 85.5 (0)      | 0.0577 |

Table 10: Certified coverage (%) of VLM filtering, CCRC (external gate) and detector-only selective prediction, and the detector’s accuracy on grounded items.

| Setting             | $\alpha = 0.10$ |      |               | $\alpha = 0.15$ |      |               | detector acc. |
|---------------------|-----------------|------|---------------|-----------------|------|---------------|---------------|
|                     | VLM filter      | CCRC | detector only | VLM filter      | CCRC | detector only |               |
| POPE-1500, LLaVA    | 68.2            | 71.1 | 57.7          | 85.5            | 91.3 | 84.8          | 83.1          |
| POPE-adv, LLaVA     | 44.5            | 48.2 | 11.6          | 72.5            | 80.5 | 43.0          | 82.4          |
| POPE-adv, Qwen2-VL  | 76.9            | 79.4 | 14.6          | 94.6            | 95.4 | 57.7          | 82.9          |
| POPE-adv, LLaVA+VCD | 45.8            | 47.8 | 14.6          | 74.4            | 76.0 | 57.7          | 82.9          |
| AMBER(d), LLaVA     | 55.2            | 56.3 | 28.3          | 74.1            | 76.1 | 45.1          | 95.6          |

of the sequence more than on the repair.

## 6.5 Where the coverage comes from

**The score.** Table 9 compares scores under the same protocol, filtering only. The scorer of ConfLVLM [41], image–text similarity, certifies 0.5% here at  $\alpha = 0.10$ ; VLM confidence alone certifies 41.0%; a learned combination of the VLM’s own signals 58.8%; adding standardised image–text similarity does not help (56.8%); adding detection grounding gives 68.2% and the lowest area under the risk–coverage curve (0.0562 against 0.0678). Against the similarity scorer the full CCRC arm (external gate, 71.1%) gains 70.6 points, of which 67.7 come from the score and 2.9 from certified repair. The correctness score is the dominant lever.

**The detector as a predictor.** If channel 2 is good, why not use it alone? Table 10 reports selective prediction on the detector alone. Its raw accuracy matches the VLM’s on POPE (83.1% against 81.9%), but its selective coverage is lower (at  $\alpha = 0.10$ , 57.7 against 68.2% on POPE-1500 and 12–28% on the other cells): accuracy does not imply a separable score.

Table 11: Self-certified repair against filtering at  $\alpha = 0.10$ : gain in points (abort rate, %) and the risk of the flip region  $r_{\text{flip}}$  (%), at both starts; the last column is CCRC with the  $q = 0.10$  detector gate.

| Setting             | $e_0$ | filter (ab) | $q=0.02$   |                   | $q=0.05$   |                   | $q=0.10$    |                   | CCRC gain |
|---------------------|-------|-------------|------------|-------------------|------------|-------------------|-------------|-------------------|-----------|
|                     |       |             | gain (ab)  | $r_{\text{flip}}$ | gain (ab)  | $r_{\text{flip}}$ | gain (ab)   | $r_{\text{flip}}$ |           |
| POPE-1500, LLaVA    | 0     | 67.5 (0)    | -53.2 (79) | 31                | -66.9 (99) | 36                | -67.5 (100) | —                 | +3.5      |
|                     | 1     | 68.2 (0)    | -35.1 (51) | 29                | -64.0 (94) | 37                | -68.0 (100) | 43                | +2.9      |
| POPE-adv, LLaVA     | 0     | 36.9 (30)   | -28.6 (84) | 54                | -34.8 (96) | 51                | -36.9 (100) | 57                | +7.1      |
|                     | 1     | 44.5 (25)   | -25.6 (68) | 50                | -39.2 (91) | 50                | -44.1 (99)  | 50                | +3.7      |
| POPE-adv, Qwen2-VL  | 0     | 76.5 (0)    | -60.0 (75) | 56                | -72.7 (94) | 53                | -76.0 (99)  | 61                | +2.7      |
|                     | 1     | 76.9 (1)    | -51.4 (63) | 55                | -70.3 (90) | 53                | -76.3 (99)  | 56                | +2.6      |
| POPE-adv, LLaVA+VCD | 0     | 39.5 (22)   | -6.8 (39)  | 12                | -27.6 (78) | 25                | -38.8 (98)  | 35                | +4.6      |
|                     | 1     | 45.8 (14)   | -4.1 (26)  | 12                | -25.7 (64) | 26                | -43.2 (96)  | 39                | +2.0      |
| AMBER(d), LLaVA     | 0     | 55.7 (0)    | -25.1 (46) | 17                | -47.3 (86) | 29                | -54.8 (98)  | 41                | -3.6      |
|                     | 1     | 55.2 (3)    | -21.4 (41) | 22                | -42.6 (78) | 32                | -52.9 (96)  | 41                | +2.5      |

**Composition with a mitigation decoder.** On POPE-adversarial with visual contrastive decoding [39] (base risk 19.7%, the highest of our cells), CCRC with the external gate gains +2.0 and +1.6 points over filtering at  $\alpha = 0.10$  and 0.15, with the sign supported by 91% and 76% of bootstrap datasets (Table 3). Lowering the base risk and adding certified repair are compatible; neither result is large.

## 6.6 Self-certified repair

A design that needs no second channel would flip the model’s answer where the score says it is probably wrong. For binary answers the flipped answer is wrong exactly when the model was right, so the flip region’s risk is the model’s *accuracy* there.

**Proposition 2** (Self-repair is bounded by slack). *Let the repair be the negation of the model’s answer on a region  $R$  of mass  $c_r$  and let the accepted region have mass  $c_a$  and risk  $r_a$ . The risk of the repaired region is  $r_r = \mathbb{P}[Y = 1 \mid X \in R]$ . If  $r_r > \alpha$ , the emitted mixture has risk at most  $\alpha$  only if  $c_r \leq c_a(\alpha - r_a)/(r_r - \alpha)$ .*

*Proof.* On  $R$  the emitted answer is  $1 - \hat{a}$ , correct iff  $\hat{a}$  is wrong, so its error indicator is  $Y$  and averaging gives  $r_r$ . The mixture risk  $(c_a r_a + c_r r_r)/(c_a + c_r) \leq \alpha$  rearranges to  $c_r(r_r - \alpha) \leq c_a(\alpha - r_a)$ .  $\square$

Table 11 flips the answer on the bottom  $q \in \{0.02, 0.05, 0.10\}$  of the score, at both starts. The flip region’s risk is 12–57%, far above  $\alpha$ , and certified coverage falls by 4 to 76 points at  $\alpha = 0.10$  with the analytic start, with abort rates up to 100%. The fixed-mass, high-error block is largest relative to the emitted set at the beginning of the sequence, fails the first hypothesis, and the sequence cannot continue to the later thresholds where Proposition 2 would allow it. The start does not change the conclusion: at  $e_0 = 0$  the losses are 7–76 points. Self-repair is not impossible in principle (the proposition admits it when the accepted region has slack and the flip region is clean), but on these cells no such region exists.

Table 12: Filtering (confidence score, image-grouped folds, 400 splits) where the detector does not apply. Certified coverage (%) with abort rate in parentheses.

| Dataset              | images | $\mu$ (%) | detector value | $\alpha=0.10$ | $\alpha=0.15$ | $\alpha=0.20$ |
|----------------------|--------|-----------|----------------|---------------|---------------|---------------|
| MME (700)            | 350    | 31.1      | 0              | 0.9 (97)      | 10.4 (56)     | 21.0 (24)     |
| GQA testdev (700)    | 319    | 27.6      | 0              | 8.5 (72)      | 34.6 (18)     | 57.7 (4)      |
| HallusionBench (447) | 165    | 48.8      | 0              | 0.0 (100)     | 0.0 (100)     | 0.0 (100)     |

## 6.7 Items without usable evidence

Table 2 shows that 0 of 1500 POPE-1500 items but 272 of 500 AMBER items have no detector value. Where the detector never applies, CCRC reduces to filtering. Table 12 gives filtering on three benchmarks where the detector returns nothing (MME, GQA, HallusionBench): at  $\alpha = 0.10$  coverage is 0.9%, 8.5% and 0% with abort rates of 97%, 72% and 100%. On HallusionBench the base risk is 48.8% and nothing is certifiable at  $\alpha \leq 0.20$ ; the floor alone is  $(0.488 - 0.20)/0.80 = 36\%$  and the abstention is far larger. Applicability of the evidence channel is a property of the benchmark’s question types, not of its difficulty.

## 6.8 Auditing validity

**Real data.** Across all arms and cells in Appendix Table 16 the test-fold exceedance of  $\alpha$  is 0.04–0.21 and the confident-violation rate is at most 0.03, against the bound of 0.10 per split. We treat this as consistent with validity and do not treat it as proof: with test folds of 150–500 items and risks near  $\alpha$ , exceedance of the realised fold risk is expected, and a real-data audit cannot exceed the resolution of its test fold.

**Known population.** The exact audit uses a generative population in which the risk of the selected policy is computed on 400,000 reference items. Items come in image clusters of six with a shared random effect that sets the intra-image correlation of errors (ICC); the score is the best item-level predictor; channel 2 is right with probability  $1 - \frac{1}{2}e^{-4v}$  at margin  $v$ . We audit the exceedance  $\mathbb{P}[\mathcal{R}(\pi_j) > \alpha]$ , counting aborts as compliant, over 3000 trials per cell,  $\mu \in \{0.15, 0.25\}$ ,  $\alpha \in \{0.10, 0.15\}$ ,  $n_{\text{cal}} \in \{150, 450\}$ .

With i.i.d. items ( $\text{ICC} = 0$ ) the exceedance is at most 0.064 in every cell for both the value-indexed family, which Theorem 1 covers, and the rank-indexed family, which it does not, so the rank-indexed variant is also conservative here and the earlier concern about it is a gap in the proof rather than an observed failure. With positive within-image correlation the guarantee degrades: at ICC 0.08 the worst cell is 0.091, and at ICC 0.26 the exceedance of the value-indexed family rises to 0.125 (filtering) and 0.132 (repair), above  $\delta$ , and the rank-indexed variant to 0.14. Calibrating on one random item per image restores validity in every cell ( $\leq 0.062$ ) at the price of a smaller calibration sample. In the benchmarks used here the measured ICC is slightly negative (Table 2), so the clustered regime is not the one these data are in; it is the regime a benchmark with several correlated questions per image would be in, and the audit shows that the item-level bound should not be used there without the one-item-per-image calibration or a design-effect correction.

Table 13: Known-population exceedance  $\mathbb{P}[\mathcal{R} > \alpha]$  (aborts counted as compliant),  $\alpha = 0.10$ ; target  $\delta = 0.10$ . “Value” fixes thresholds and gate from an independent fit sample (the object of Theorem 1); “rank” takes them as quantiles of the calibration sample; the last column calibrates on one random item per image.

| ICC        | $\mu$ | $n_{\text{cal}}$ | value, filter | value, repair | rank, filter | rank, repair | value, repair, 1 item/image |
|------------|-------|------------------|---------------|---------------|--------------|--------------|-----------------------------|
| 0 (i.i.d.) | 0.15  | 150              | 0.052         | 0.063         | 0.049        | 0.064        | —                           |
| 0 (i.i.d.) | 0.15  | 450              | 0.063         | 0.051         | 0.051        | 0.058        | —                           |
| 0 (i.i.d.) | 0.25  | 150              | 0.046         | 0.052         | 0.036        | 0.046        | —                           |
| 0 (i.i.d.) | 0.25  | 450              | 0.047         | 0.050         | 0.058        | 0.055        | —                           |
| 0.08       | 0.15  | 150              | 0.060         | 0.071         | 0.063        | 0.076        | 0.053                       |
| 0.08       | 0.15  | 450              | 0.065         | 0.078         | 0.090        | 0.067        | 0.051                       |
| 0.08       | 0.25  | 150              | 0.050         | 0.068         | 0.053        | 0.067        | 0.045                       |
| 0.08       | 0.25  | 450              | 0.054         | 0.067         | 0.056        | 0.078        | 0.052                       |
| 0.26       | 0.15  | 150              | 0.078         | 0.102         | 0.069        | 0.103        | 0.041                       |
| 0.26       | 0.15  | 450              | 0.068         | 0.111         | 0.081        | 0.110        | 0.056                       |
| 0.26       | 0.25  | 150              | 0.027         | 0.080         | 0.031        | 0.099        | 0.042                       |
| 0.26       | 0.25  | 450              | 0.043         | 0.085         | 0.043        | 0.081        | 0.048                       |

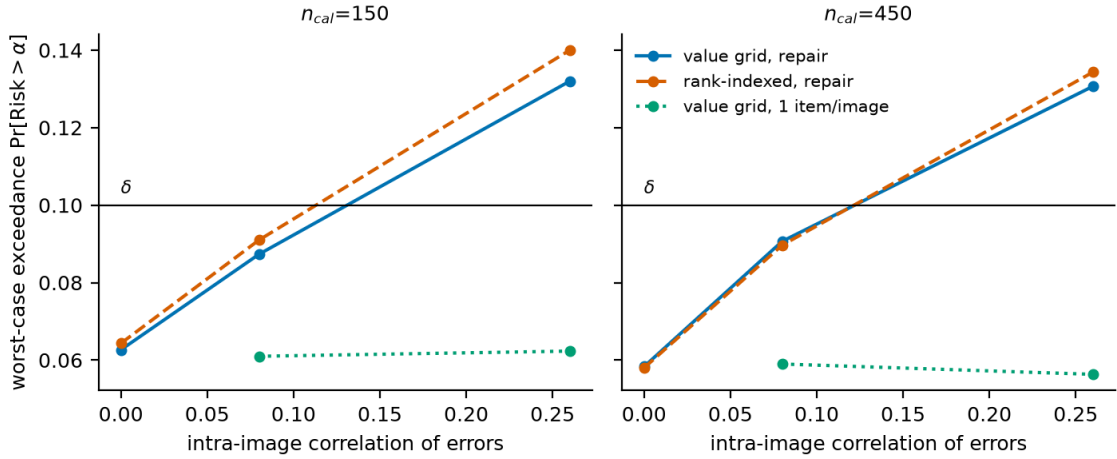


Figure 4: Worst-case known-population exceedance over the  $\mu$  and  $\alpha$  cells, against the intra-image correlation of errors, with repair. Horizontal line:  $\delta = 0.10$ .

## 7 Comparison with concurrent methods

Two methods published in 2026 are close enough to CCRC that a reader will ask what separates them. Table 14 compares the three on the dimensions the reviewing literature cares about. The comparison is conceptual: the benchmarks, backbones and sample sizes differ, so we do not compare numbers across papers and we did not rerun either method.

**BCEA.** BCEA reduces the abstention floor’s practical bite by spending a visual budget to re-read the image and recalibrating on post-acquisition scores. It stays inside the premise of Proposition 1: acquisition changes how confident the method is about the model’s claim, not which claim is emitted. Its current version certifies a Bonferroni-corrected Clopper–Pearson test over a threshold grid fixed independently of the calibration sample and reports certified coverage of 0.23 without acquisition and 0.32 with gated acquisition at  $\alpha = 0.10$ ; the coverage figures of

Table 14: CCRC, BCEA [71] (version 4, 21 July 2026) and CEBC [49].

|                                     | CCRC                                                                     | BCEA                                                                                                             | CEBC                                                                                                     |
|-------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Action on a flagged claim           | emit a detector answer, accept, or abstain                               | acquire more visual evidence, then answer or abstain; the answer is the model’s own                              | rewrite with a VLM, possibly substituting or inserting objects, then delete what remains below threshold |
| Quantity controlled                 | error rate of the emitted set, accepted and substituted answers together | error rate of emitted claims                                                                                     | per-mention false-accept rate of a detector threshold on hallucinated mentions in base captions          |
| Guarantee                           | finite-sample, probability $1 - \delta$ over calibration, i.i.d. units   | finite-sample, Bonferroni-corrected Clopper–Pearson over a fixed grid; a full-level scan is labelled uncertified | none stated; no theorem, no sampling model                                                               |
| Revised output inside the guarantee | yes                                                                      | not applicable (no revision)                                                                                     | no; the rewrite may use objects at $\tau - 0.05$ and is outside the calibration                          |
| Calibration procedure               | fixed-sequence test over values from the fit fold                        | fixed grid, Bonferroni                                                                                           | one clipped quantile                                                                                     |
| Role of the detector                | channel 2: supplies the substituted answer and its margin                | optional guide for acquisition                                                                                   | sole arbiter of support                                                                                  |

0.28 and 0.37 correspond to its uncertified practical variant, which scans the calibration scores at full confidence. The mechanisms are different (re-scoring a claim versus replacing it) and in principle composable, since a re-scored claim can still be repaired when the detector is decisive. We have not tested the composition in this version, and nothing here shows that it helps.

**CEBC.** CEBC is the nearest published method that edits unsupported object mentions in VLM output with an external detector. Its calibration is a one-sided quantile of detector scores on hallucinated mentions from base captions; its rewrite stage is not calibrated; and it reports no violation check. Whatever its empirical merit, it is a detector-gated rewriter rather than a certified selective predictor, and its headline numbers (for example CHAIR<sub>S</sub> from 10.0 to 2.0) measure a different object than a certified error rate over emitted claims. CEBC applies to free-form captions and CCRC as evaluated to yes/no questions; the guarantee of Theorem 1 has no free-form counterpart in this paper, and §8 gives the reason.

**What remains of the novelty claim.** Among the work we surveyed up to 4 October 2026, CCRC is the one whose certificate covers an emitted answer that differs from the model’s, with the repair gate fixed from fit or external data. CEBC edits without certifying; BCEA certifies without editing; conformal linguistic calibration [31] certifies a rewritten output for text-only question answering by generalising claims. The claim is about the combination, in the setting of object-existence questions, and not about the idea of correcting hallucinations.

Table 15: Paired matched-prefix test ( $n = 121$ ). All  $p$ -values are two-sided.

| Downstream measure (continuation only) | Original | Repaired | paired difference ( $p$ ) |
|----------------------------------------|----------|----------|---------------------------|
| Hallucinated mentions (mean)           | 1.207    | 1.413    | +0.207 (0.025)            |
| Hallucinations per 100 words           | 3.61     | 4.30     | +0.75 (0.011)             |
| Faithful mentions (mean)               | 1.579    | 1.752    | +0.174 (0.077)            |
| Continuation length (words)            | 33.4     | 32.9     | −0.55 (0.092)             |
| Hallucinated share of mentions         | 0.433    | 0.447    | −0.004 (0.884)            |

## 8 Free-form generation: mid-sequence repair increases downstream hallucination

Correction looks more attractive in captions than in yes/no answers: replacing a hallucinated word keeps a caption that deletion would damage. The extension seems mechanical, and two obstacles are known. Editing a prefix makes later tokens off-policy, which is feedback covariate shift [18] and falls outside weighted conformal methods [59]; and per-claim guarantees say nothing about a whole sequence whose length is data-dependent. Prefix-conditional (Mondrian)  $p$ -values and e-processes [54, 67] could handle both in principle, leaving the drift of the calibration pool, which on-policy calibration by fixed-point iteration would address if the correction map were monotone. We measured whether it is before attempting a proof.

**Design.** For each image we generate a caption greedily, locate the first hallucinated object mention at position  $t$  using AMBER’s per-image present and absent object lists, and build two prefixes identical up to  $t$ : one that keeps the hallucinated word and one that substitutes the best-detected object the image contains. We continue decoding the same number of tokens from each and count hallucinated mentions in the continuations only. The only difference between the two continuations is the intervention. The unit is the matched pair ( $n = 121$  images); there is one repair per pair.

**Result.** Repair makes the continuation worse (Table 15, Figure 5). Hallucinated mentions in the continuation rise from 1.207 to 1.413, a paired difference of  $+0.207 \pm 0.091$  ( $t(120) = 2.27$ , two-sided  $p = 0.025$ ), with 45 pairs worsening, 25 improving and 51 tied. A Wilcoxon signed-rank test ( $p = 0.035$ ) and an exact sign test ( $p = 0.023$ ) agree. The Hodges–Lehmann estimate is 0.000 because 42% of pairs are tied.

**What the experiment can support.** The result is significant under three tests and underpowered. At the observed paired standard deviation (0.999) the minimum detectable effect at 80% power is +0.257, larger than the observed +0.207, and about 186 pairs would be needed at this effect size. Equivalence testing against a margin of  $\pm 0.25$  does not establish equivalence ( $p = 0.317$ ). The sign is established and the magnitude is not: any value in  $[+0.027, +0.386]$  is compatible. The number of hallucinated mentions rises together with faithful mentions and with mention density; per mention, the hallucinated share does not move ( $0.433 \rightarrow 0.447$ ,  $p = 0.884$ ), so the mechanism is denser object talk after repair. A certificate on the count of false claims is not helped by this, since each additional hallucinated mention counts. The substitution used

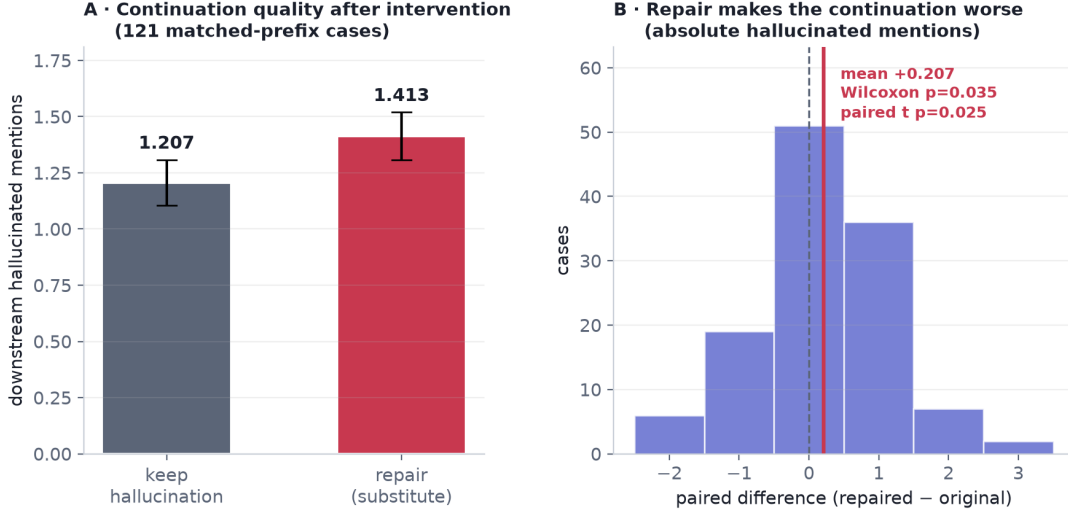


Figure 5: (A) Mean hallucinated mentions in the continuation after keeping or repairing the flagged claim (standard errors). (B) Distribution of the paired difference.

here is context-blind, and the finding concerns that rule rather than all correction. What the data withdraw is support for a monotone correction map and hence for a fixed-point route to a sequential guarantee, at the sample size we have.

**A coupling bound.** For a sequence of claims let  $F(O)$  be the number of false claims and  $\mathcal{R}(O) = \mathbb{E}[F(O)]$  the expected count per sequence. Let a repair rule replace claims at a random set  $P$  of positions, producing  $O'$ . Write  $W$  for the number of wrong repaired claims,  $F_P(O)$  for the number of false *original* claims at the repaired positions, and  $\Delta$  for the signed change in the number of false claims at non-repaired positions between  $O'$  and  $O$ .

**Theorem 2** (Coupling bound for corrected sequences).  $F(O') = F(O) - F_P(O) + W + \Delta$  pointwise, hence

$$\mathcal{R}(O') = \mathcal{R}(O) + \mathbb{E}[W] - \mathbb{E}[F_P(O)] + \mathcal{I} \leq \mathcal{R}(O) + \mathbb{E}[W] + \mathcal{I}, \quad \mathcal{I} := \mathbb{E}[\Delta].$$

$\mathcal{I}$  is signed: a repair that prevents later errors makes it negative. If at most one claim is repaired per sequence,  $\mathbb{E}[W] = \mathbb{P}[X \in R] \mathbb{P}[\text{repair wrong} \mid X \in R]$ ; in general  $\mathbb{E}[W] = \sum_t \mathbb{P}[t \in P, \text{repair at } t \text{ wrong}]$  is an expected count over repair opportunities.

*Proof.* The false claims of  $O'$  are those at repaired positions (the wrong repairs,  $W$ ) plus those at non-repaired positions. The false claims of  $O$  are those at repaired positions ( $F_P(O)$ ) plus those at non-repaired positions. Subtracting gives the identity, and taking expectations gives the equality;  $F_P(O) \geq 0$  gives the bound.  $\square$

In the matched-prefix experiment one claim is repaired per pair, so  $\hat{\mathcal{I}}$  is the paired difference in the continuation:  $+0.207 \pm 0.091$  additional false claims per repair. That is as large as the direct term for a repair that is wrong with probability 0.2, and it cannot be dropped. Certifying  $\mathcal{R}(O) \leq \alpha_1$  and the direct term by  $\alpha_2$  at  $\delta/2$  each gives  $\mathcal{R}(O') \leq \alpha_1 + \alpha_2 + \mathcal{I}$  at  $1 - \delta$ , which is useful only relative to an already-selective base policy, since certifying the uncorrected generation

is the object the abstention floor makes expensive. A sequential procedure would have to price the continuation and not only the claim, for example by searching over rewrites under an accumulated risk ledger. We leave that open.

## 9 Discussion

**What the guarantee says.** With probability  $1 - \delta$  over the calibration sample, the population risk of the deployed policy, accepted and substituted answers together, is at most  $\alpha$ . It says nothing about a single item, about a finite test fold, about a distribution other than the one the units were drawn from, or about the coverage that results. A sampling model is part of it: units are i.i.d., and when a benchmark is split at random the iid assumption is a modelling choice about the population the benchmark represents.

**What the evidence supports.** Under a protocol in which the proved procedure is the evaluated procedure, certified repair gains between 0.8 and 7.9 points of coverage over certified filtering when the gate is fixed on external data; the sign is supported by 65–99% of image-bootstrap datasets and the interval excludes zero on one cell. The statement a reader can rely on is therefore that the gain is positive in expectation on these benchmarks and small, and that more images would be needed to size it. Two of the three mechanisms the earlier analysis offered survive in weaker form. Repair does add a few points of coverage; it does so mostly by confirming answers the model already gave, and corrects almost none of the model’s false positives. The correctness score dominates the coverage, and the start of the fixed sequence dominates the repair on small folds.

**Limitations.** (i) The task is binary object existence, with one detector, three backbones and two benchmark families whose images overlap; 642 distinct images are involved in total. (ii) The channel-2 operating point  $\theta = 0.15$  was fixed in the extraction scripts and is not varied here; we do not claim it was chosen independently of the benchmarks. (iii) The external gate for the POPE cells was chosen on AMBER, whose image sources we could not verify to be disjoint from POPE’s COCO images; the guarantee requires the gate to be independent of the calibration sample, and any image overlap would weaken that independence slightly. (iv) The i.i.d. assumption is violated within images when errors are correlated; here the measured correlation is slightly negative, and §6.8 shows the guarantee degrades with positive correlation unless calibration uses one item per image. (v) The bootstrap interval reflects sampling of images from the benchmark, not of benchmarks from the world. (vi) Free-form generation is not covered and mid-sequence repair worsens the continuation. (vii) We did not rerun the VLMs, extract signals on additional images, or run BCEA or CEBC.

**Next steps.** The measurement that would most improve the paper is more images: processing the remaining POPE images and a second detector would narrow the intervals the bootstrap now shows to be wide. A repair rule that respects the generation context, tested on the free-form setting with the matched-prefix design, is the natural extension.

## 10 Conclusion

Certified correction replaces an answer the model would have abstained on with a detector’s answer and certifies the emitted set as a whole; the procedure is valid under i.i.d. sampling when thresholds and gate are fixed before calibration. Under a protocol built on that requirement the coverage gain over filtering is small and positive, with an uncertainty that the available images do not resolve. Most repaired items re-emit the model’s answer with detector confirmation, the model’s false positives are not corrected, and the coverage comes mainly from the correctness score. In generation, mid-sequence repair worsens the continuation. The reproducible pipeline and all negative results are released.

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## A Additional results

### A.1 Audit of all arms

Table 16: Certified coverage (%) with abort rate (ab) and validity statistics for every arm, all items. exc is the fraction of non-aborted splits whose test-fold risk exceeds  $\alpha$ ; cv is the fraction whose one-sided 90% Clopper–Pearson lower bound on test risk exceeds  $\alpha$ . The fixed  $q=0.10$  gate was chosen with knowledge of these benchmarks and is not certified.

| Setting             | $\alpha$ | Filter    |             | External gate |             | Fit-fold gate |             | Union ( $\delta/3$ ) |             | Fixed $q=0.10$ |             |
|---------------------|----------|-----------|-------------|---------------|-------------|---------------|-------------|----------------------|-------------|----------------|-------------|
|                     |          | cov. (ab) | exc/cv      | cov. (ab)     | exc/cv      | cov. (ab)     | exc/cv      | cov. (ab)            | exc/cv      | cov. (ab)      | exc/cv      |
| POPE-1500, LLaVA    | 0.10     | 68.2 (0)  | 0.11 / 0.02 | 71.1 (0)      | 0.11 / 0.01 | 77.0 (1)      | 0.10 / 0.01 | 72.6 (0)             | 0.04 / 0.01 | 71.1 (0)       | 0.11 / 0.01 |
|                     | 0.15     | 85.5 (0)  | 0.09 / 0.01 | 91.3 (0)      | 0.10 / 0.01 | 92.6 (0)      | 0.09 / 0.01 | 89.1 (0)             | 0.04 / 0.00 | 87.6 (0)       | 0.09 / 0.01 |
| POPE-adv, LLaVA     | 0.10     | 44.5 (25) | 0.14 / 0.01 | 48.2 (22)     | 0.14 / 0.02 | 41.4 (34)     | 0.14 / 0.00 | 28.0 (57)            | 0.12 / 0.01 | 48.2 (22)      | 0.14 / 0.02 |
|                     | 0.15     | 72.5 (2)  | 0.07 / 0.00 | 80.5 (2)      | 0.07 / 0.00 | 82.0 (2)      | 0.08 / 0.00 | 72.7 (4)             | 0.04 / 0.00 | 76.4 (1)       | 0.09 / 0.00 |
| POPE-adv, Qwen2-VL  | 0.10     | 76.9 (1)  | 0.09 / 0.01 | 79.4 (0)      | 0.10 / 0.01 | 71.2 (10)     | 0.11 / 0.01 | 68.9 (11)            | 0.05 / 0.00 | 79.4 (0)       | 0.10 / 0.01 |
|                     | 0.15     | 94.6 (0)  | 0.06 / 0.00 | 95.4 (0)      | 0.06 / 0.00 | 95.0 (0)      | 0.06 / 0.00 | 92.9 (0)             | 0.04 / 0.00 | 95.2 (0)       | 0.06 / 0.00 |
| POPE-adv, LLaVA+VCD | 0.10     | 45.8 (14) | 0.14 / 0.01 | 47.8 (11)     | 0.14 / 0.01 | 39.6 (30)     | 0.21 / 0.03 | 34.2 (37)            | 0.10 / 0.01 | 47.8 (11)      | 0.14 / 0.01 |
|                     | 0.15     | 74.4 (0)  | 0.11 / 0.02 | 76.0 (1)      | 0.11 / 0.02 | 76.1 (0)      | 0.11 / 0.02 | 69.4 (0)             | 0.04 / 0.01 | 75.7 (0)       | 0.11 / 0.02 |
| AMBER(d), LLaVA     | 0.10     | 55.2 (3)  | 0.10 / 0.02 | 56.3 (6)      | 0.09 / 0.02 | 55.8 (6)      | 0.10 / 0.02 | 41.3 (29)            | 0.06 / 0.00 | 57.7 (4)       | 0.09 / 0.01 |
|                     | 0.15     | 74.1 (0)  | 0.15 / 0.03 | 76.1 (0)      | 0.13 / 0.03 | 75.3 (0)      | 0.13 / 0.03 | 71.1 (0)             | 0.06 / 0.01 | 75.5 (1)       | 0.14 / 0.03 |

### A.2 Protocol ladder and start of the sequence at other targets

Table 17: As Table 4 at  $\alpha = 0.15$ .

|                                                                                      | POPE-1500, LLaVA | POPE-adv, LLaVA | POPE-adv, Qwen2-VL | POPE-adv, LLaVA+VCD | AMBER(d), LLaVA |
|--------------------------------------------------------------------------------------|------------------|-----------------|--------------------|---------------------|-----------------|
| conventional protocol (item folds, calibration quantiles, $e_0=0$ , fixed $q=0.10$ ) | 86.4/88.4        | 69.3/76.7       | 94.8/95.2          | 75.8/77.2           | 95.9/84.0       |
| + image-grouped folds                                                                | 85.8/87.8        | 70.4/77.7       | 94.5/94.9          | 75.6/76.8           | 96.3/88.4       |
| + fixed value grid and gate (fit fold)                                               | 85.5/87.5        | 61.8/72.9       | 93.7/94.7          | 63.1/72.4           | 94.9/94.1       |
| + analytic start $e_0=1$                                                             | 85.5/87.6        | 70.9/75.1       | 94.1/94.9          | 73.5/75.1           | 95.0/95.0       |
| + gate chosen on the fit fold                                                        | 85.5/92.6        | 70.9/81.1       | 94.1/94.7          | 73.5/75.5           | 95.0/95.1       |
| + all-item cohort                                                                    | 85.5/92.6        | 72.5/82.0       | 94.6/95.0          | 74.4/76.1           | 74.1/75.3       |

Table 18: As Table 8 at  $\alpha = 0.05$ .

| $c$ | $e_0$ | $k_{\text{start}}$ | POPE-1500, LLaVA |           | POPE-adv, LLaVA |           | POPE-adv, Qwen2-VL |           | POPE-adv, LLaVA+VCD |           | AMBER(d), LLaVA |           |
|-----|-------|--------------------|------------------|-----------|-----------------|-----------|--------------------|-----------|---------------------|-----------|-----------------|-----------|
|     |       |                    | filter           | CCRC      | filter          | CCRC      | filter             | CCRC      | filter              | CCRC      | filter          | CCRC      |
| 1.0 | 0     | 45                 | 19.6 (35)        | 29.2 (18) | 8.6 (80)        | 7.9 (82)  | 35.4 (17)          | 36.7 (19) | 9.8 (74)            | 9.6 (75)  | 17.6 (50)       | 20.6 (46) |
| 1.0 | 1     | 77                 | 27.5 (14)        | 31.4 (16) | 3.2 (94)        | 3.2 (94)  | 31.6 (46)          | 29.6 (50) | 8.7 (82)            | 8.7 (82)  | 6.1 (88)        | 6.3 (88)  |
| 1.0 | 2     | 105                | 24.2 (32)        | 27.2 (30) | 0.5 (99)        | 0.5 (99)  | 37.1 (40)          | 34.3 (46) | 2.2 (96)            | 2.2 (96)  | 0.0 (100)       | 0.0 (100) |
| 1.0 | 3     | 132                | 21.2 (50)        | 22.2 (51) | 0.0 (100)       | 0.0 (100) | 7.4 (90)           | 7.3 (90)  | 0.3 (100)           | 0.3 (100) | 0.0 (100)       | 0.0 (100) |
| 1.5 | 0     | 45                 | 26.3 (16)        | 31.6 (15) | 4.7 (91)        | 4.6 (91)  | 30.4 (44)          | 26.6 (52) | 8.2 (82)            | 8.2 (82)  | 9.1 (80)        | 8.8 (80)  |
| 1.5 | 1     | 77                 | 21.8 (44)        | 24.3 (41) | 0.0 (100)       | 0.0 (100) | 24.1 (62)          | 23.4 (64) | 1.2 (98)            | 1.2 (98)  | 0.0 (100)       | 0.0 (100) |
| 1.5 | 2     | 105                | 23.1 (50)        | 24.6 (48) | 0.0 (100)       | 0.0 (100) | 1.6 (98)           | 1.6 (98)  | 0.0 (100)           | 0.0 (100) | 0.0 (100)       | 0.0 (100) |
| 1.5 | 3     | 132                | 25.0 (49)        | 25.8 (48) | 0.0 (100)       | 0.0 (100) | 0.0 (100)          | 0.0 (100) | 0.0 (100)           | 0.0 (100) | 0.0 (100)       | 0.0 (100) |

Table 19: As Table 8 at  $\alpha = 0.15$ .

| $c$ | $e_0$ | $k_{\text{start}}$ | POPE-1500, LLaVA |          | POPE-adv, LLaVA |          | POPE-adv, Qwen2-VL |          | POPE-adv, LLaVA+VCD |          | AMBER(d), LLaVA |          |
|-----|-------|--------------------|------------------|----------|-----------------|----------|--------------------|----------|---------------------|----------|-----------------|----------|
|     |       |                    | filter           | CCRC     | filter          | CCRC     | filter             | CCRC     | filter              | CCRC     | filter          | CCRC     |
| 1.0 | 0     | 15                 | 85.5 (0)         | 92.6 (0) | 46.1 (36)       | 81.9 (1) | 91.9 (3)           | 92.6 (2) | 38.9 (47)           | 73.4 (4) | 73.6 (0)        | 73.0 (3) |
| 1.0 | 1     | 25                 | 85.5 (0)         | 92.6 (0) | 67.6 (8)        | 82.9 (0) | 94.3 (0)           | 94.9 (0) | 66.2 (11)           | 74.7 (2) | 73.9 (0)        | 74.3 (2) |
| 1.0 | 2     | 34                 | 85.6 (0)         | 92.6 (0) | 71.4 (3)        | 82.4 (1) | 94.3 (0)           | 94.8 (0) | 73.2 (2)            | 75.2 (1) | 74.1 (0)        | 74.9 (1) |
| 1.0 | 3     | 43                 | 85.6 (0)         | 92.6 (0) | 73.2 (1)        | 81.3 (2) | 94.5 (0)           | 94.9 (0) | 74.4 (0)            | 75.6 (0) | 74.1 (0)        | 75.5 (0) |
| 1.5 | 0     | 15                 | 85.5 (0)         | 92.6 (0) | 64.0 (12)       | 82.4 (0) | 94.3 (0)           | 94.8 (0) | 61.7 (18)           | 74.0 (3) | 74.1 (0)        | 74.4 (2) |
| 1.5 | 1     | 25                 | 85.5 (0)         | 92.6 (0) | 72.5 (2)        | 82.0 (2) | 94.6 (0)           | 95.0 (0) | 74.4 (0)            | 76.1 (0) | 74.1 (0)        | 75.3 (0) |
| 1.5 | 2     | 34                 | 85.5 (0)         | 92.6 (0) | 72.6 (3)        | 81.3 (3) | 94.6 (0)           | 95.2 (0) | 74.5 (0)            | 76.1 (0) | 74.2 (0)        | 75.5 (0) |
| 1.5 | 3     | 43                 | 85.6 (0)         | 92.6 (0) | 72.8 (4)        | 79.9 (5) | 94.8 (0)           | 95.1 (0) | 74.6 (1)            | 74.6 (3) | 74.1 (0)        | 75.7 (0) |

Table 20: As Table 3 restricted to items with a detector value; no bootstrap.

| Setting             | $\alpha$ | Filter    | External gate |      | Fit-fold gate |       | Fixed $q=0.10$ |      |
|---------------------|----------|-----------|---------------|------|---------------|-------|----------------|------|
|                     |          | cov. (ab) | cov. (ab)     | gain | cov. (ab)     | gain  | cov. (ab)      | gain |
| POPE-1500, LLaVA    | 0.10     | 68.2 (0)  | 71.1 (0)      | +2.9 | 77.0 (1)      | +8.8  | 71.1 (0)       | +2.9 |
|                     | 0.15     | 85.5 (0)  | 91.3 (0)      | +5.8 | 92.6 (0)      | +7.1  | 87.6 (0)       | +2.1 |
| POPE-adv, LLaVA     | 0.10     | 43.6 (26) | 47.4 (22)     | +3.8 | 39.4 (37)     | -4.2  | 47.4 (22)      | +3.8 |
|                     | 0.15     | 70.9 (2)  | 79.3 (2)      | +8.3 | 81.1 (1)      | +10.2 | 75.1 (1)       | +4.2 |
| POPE-adv, Qwen2-VL  | 0.10     | 75.6 (1)  | 78.0 (1)      | +2.5 | 68.3 (13)     | -7.3  | 78.0 (1)       | +2.5 |
|                     | 0.15     | 94.1 (0)  | 95.0 (0)      | +0.9 | 94.7 (0)      | +0.6  | 94.9 (0)       | +0.8 |
| POPE-adv, LLaVA+VCD | 0.10     | 45.5 (15) | 47.6 (12)     | +2.1 | 39.7 (30)     | -5.8  | 47.6 (12)      | +2.1 |
|                     | 0.15     | 73.5 (2)  | 75.8 (2)      | +2.3 | 75.5 (2)      | +2.0  | 75.1 (1)       | +1.6 |
| AMBER(d), LLaVA     | 0.10     | 90.2 (1)  | 88.6 (6)      | -1.7 | 89.8 (4)      | -0.5  | 90.0 (4)       | -0.3 |
|                     | 0.15     | 95.0 (0)  | 95.8 (0)      | +0.8 | 95.1 (0)      | +0.1  | 95.0 (0)       | +0.1 |

### A.3 Grounded-only cohort

### A.4 Reproducibility

All numbers in the paper are generated from per-item signals by the scripts in the released repository (`conformal_followup/revision`): `attach_image_ids.py` recovers and verifies image identifiers; `ccrc_rev.py` implements the protocol; `regress_legacy.py` checks that the conventional-protocol mode reproduces the numbers of the baseline implementation; `exp_*.py` produce the results files and `make_tables.py`, `make_figures.py` the tables and figures. Splits are deterministic in the seed.